On Various Integral or Summatory Transformations of Functions

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1 Arithmetic Sub-series of the Taylor Expansion

Suppose $f(x) = \sum_n a_n x^n$ and define the sub-series by selecting every kth term; $f_k(x) = \sum_n a_{nk} x^{nk}$. We of course have;

$$f_k(x) = \frac{1}{k} \sum_{j=0}^{k-1} f\left(x \cdot e^{\frac{2\pi i j}{k}}\right)$$

Occasionally we also see more general examples of this effect, for example:

$$\sum_{k=1}^{\infty} \frac{x^{2p}}{k^{2p} - x^{2p}} = \sum_{m=1}^{\infty} x^{2pm} \zeta(2pm) = \left(\frac{1 - \pi x \cot(\pi x)}{2}\right)_p = \left(\sum_{k=1}^{\infty} \frac{x^2}{k^2 - x^2}\right)_p$$

Here $()_k$ implies the weighted sum over roots of unity stated above. This sifting property of the roots of unity for sums of $c_{an+b} x^{an+b}$ gives all sorts of intriguing results when applied to common functions, such as the logarithm below;

$$\sum_{n=0}^{\infty} \frac{x^{(an+b)}}{an+b} = -\frac{1}{a} \sum_{j=0}^{a-1} \left(\frac{1}{2} \cos\left(\frac{2\pi b}{a} j\right) \ln\left(x^2 + 1 - 2x\cos\left(\frac{2\pi}{a} j\right)\right) + \sin\left(\frac{2\pi b}{a} j\right) \arctan\left(\frac{-x \sin\left(\frac{2\pi}{a} j\right)}{1 - x\cos\left(\frac{2\pi}{a} j\right)}\right)\right)$$

But on the hunt for *integral* transforms, it is straightforward to note that the following operation on f achieves the sub-series function;

$$\sum_n a_{(an+b)} x^{(an+b)} = \frac{1}{2\pi i} \int_{C_x} \frac{f(z)}{z} \left(\frac{x}{z}\right)^b \frac{1}{1 - \left(\frac{x}{z}\right)^a} dz$$

Where C_x is chosen to include the disc of radius x, ideally excluding any poles of f. Let's examine an application with the entire function e^z and the Mittag-Leffler function $E_{a,b}$ for integers a, b and real x;

$$E_{a,b}(x) = \sum_{n \geq 0} \frac{x^n}{\Gamma(an+b)} = \frac{x^{\frac{(1-b)}{a}}}{a} \sum_{m=0}^{a-1} e^{x^{\frac{1}{a}} \cos\left(\frac{2\pi m}{a}\right)} \cos\left(x^{\frac{1}{a}} \sin\left(\frac{2\pi m}{a}\right) - \frac{(b-1)\, 2\pi m}{a}\right) \; ; \; 1 \leq b \leq a$$

As is seen by computing

$$\sum_{m=0}^{a-1} \frac{e^{x w_m}}{x w_m} \left(\frac{x}{x w_m}\right)^b \cdot \lim_{z \to w_m} \frac{(z - x w_m)}{1 - x^a z^{-a}}$$

The interested reader will find that to generate a *geometric* subseries of a Taylor Expansion, a Dirichlet convolution with the Mobius function, $\mu(rn)$, gives an indicator for powers of r.

2 Transforms of Log Gamma

Building off these ideas of coupling Laplace transforms and recursion, here is a related result;

$$\mathcal{L}\{\ln \Gamma(x)\} = e^s \int_1^{\infty} e^{-st} \ln(\Gamma(t))\, dt - \int_0^{\infty} e^{-st} \ln(t)\, dt$$

$$= \frac{-e^s \int_0^1 e^{-st} \ln\left(\Gamma\left(t\right)\right) dt + \frac{\gamma+\ln(s)}{s}}{1-e^s}$$

To solve the remaining integral, we employ Kummer's Fourier series for log Gamma, and the Fourier series of $\log\sin(x)$

$$\int_0^1 e^{-sx} \left(\left(\frac{1}{2}-x\right)(g+\ln(2)) + (1-x)\ln(\pi) - \frac{1}{2}\ln(\sin(\pi x)) + \frac{1}{\pi}\sum_{n=1}^{\infty} \frac{\ln(n)}{n}\sin(2\pi n x) \right) dx$$

With integration by parts;

$$\mathcal{L} = \frac{1}{1-e^s}\left(-e^s\left((\gamma+\ln(2))\left(\frac{e^{-s}\left(e^s(s-2)+s+2\right)}{2s^2}\right) + \ln(\pi)\left(\frac{s+e^{-s}-1}{s^2}\right) - \frac{1}{2}\left(-\sum_{k=1}^{\infty}\frac{1}{k}\left(\frac{e^{-s}\left(-s+e^s s\right)}{4\pi^2k^2+s^2}\right) - \ln(2)\cdot\frac{e^{-s}-1}{-s} + \sum_{n=1}^{\infty}\ln(n)\left(\frac{e^{-s}\left(2e^s-2\right)}{4\pi^2n^2+s^2}\right) + \frac{\gamma+\ln(s)}{s}\right.\right.\right.$$

$$\boxed{\frac{-\gamma}{2s} + (\gamma+\ln(2\pi))\frac{s-1}{s^2} + \frac{1}{s\left(1-e^s\right)}\ln\left(\frac{s}{2\pi}\right) + \sum_{k=1}^{\infty}\frac{1}{4\pi^2k^2+s^2}\left(\frac{1}{k}\frac{s}{2} + 2\ln(k)\right)}$$

Interestingly these summands are expressible in terms of two non-standard Zeta generating functions;

$$\sum_{n=0}^{\infty} x^{2n}\zeta'(2n)$$

And

$$\sum_{n=0}^{\infty} x^{2n}\zeta(2n-1)$$

Taking a less rigorous approach, suppose we desire the inverse Laplace Transform to log Gamma which satisfies

$$l(s+1) = l(s) + \ln(s) \rightarrow e^{-t} f(t) = f(t) + \mathcal{L}^{-1}\left\{\ln(s)\right\}$$

Where the inverse Laplace transform of ln has but a divergent principal value of $-\frac{1}{t}L^{-1}\left\{\frac{d}{ds}\ln(s)\right\} = -\frac{1}{t}$. Putting these expressions together, we have the questionable

$$L^{-1}\left\{\ln\Gamma(s)\right\} = \frac{-\frac{1}{t}}{e^{-t}-1} + \delta(t)h(t)$$

For a particular $h(t)$. More productively, we obtain from this;

$$L^{-1}\left\{\frac{d^{m+1}}{ds^{m+1}}\ln\Gamma(s)\right\} = \frac{(-1)^{m+1}t^m}{1-e^{-t}} \rightarrow \psi_m(z) = \int_0^{\infty}\frac{e^{-zt}(-t)^m}{e^{-t}-1}dt$$

A standard result. Taking a quick integral gives Gauss' elementary Digamma integral;

$$\int_1^x\int_0^{\infty}\frac{e^{-zt}(-t)}{e^{-t}-1}dtdz + \psi(1) = \int_0^{\infty}\left(\frac{e^{-xt}}{e^{-t}-1} + \frac{e^{-t}}{t}\right)dt$$

3 Loose Spherical Vector Calculus Relationships

Spheres, surface integrals, Fourier transforms, and homogeneous functions all have a nice interplay as a result of the fact that the sphere is uniform under scaling and has normal vectors aligned with the radius vector.

$$\iint_{\partial B(0,R)} F\left\{Div\left\{A\right\}\right\} ds \rightarrow \iint_{\partial B(0,R)} iw\cdot F\left\{A\right\} ds \rightarrow i\iint_{\partial B(0,R)} n\cdot F\left\{A\right\} ds \rightarrow i\iiint_{B(0,R)} Div\left\{F\left\{A\right\}\right\} dV$$

In this manner, the multidimensional Fourier transform has passed through the derivative operator. We also have the nice;

$$\iint_{\partial B(0,R)} F\left\{Curl\left\{A\right\}\right\}\cdot nds = i\iint_{\partial B(0,R)} w\times F\left\{A\right\}\cdot nds = 0$$

Akin to Harmonic functions, Homogeneous functions also have constrained integrals over Balls in $\mathcal{R}^n$. With homogeneous scalar f of order k we have simply;

$$kR^{n-2+k}\int_{\partial B(0,1)} fd\sigma = \int_{B(0,R)} \nabla^2 fdV$$

Further;

$$\phi\left(r\right) = \int_{dB(0,r)} f\left(x\right)d\sigma \to \phi\left(sr\right) = s^{n+k-1}\phi\left(r\right)$$

Alongside Euler's Theorem for homogeneous functions, even more vector calculus identities can be mixed into such a relationship. We even have an umbral-flavored 'fundamental theorem' under usage of the sphere-normal trick;

$$\int_0^R \phi_{\nabla^2 f}\left(r\right)dr = \int_{B(0,R)} \nabla\cdot\nabla fdV = \int_{dB(0,R)} \nabla f\cdot nd\sigma = \int_{dB(0,R)} \nabla f\left(r\right)\cdot\frac{r}{R}d\sigma = \frac{k}{R}\int_{dB(0,R)} fd\sigma = \frac{k}{R}\left[\phi_f\left(R\right) - \phi_f\left(0\right)\right]$$

More of a party trick than anything, but demonstrating a kind of $\phi_{\nabla^2 f} \to \phi'$ correspondence.

3.1 Vectors onto Matrices

If instead of integral transforms on vector functions we consider the discrete sum arising from matrix-vector multiplication, we find an interesting analogy;

$$Av = \sum_j A_{ij}v_j \ \to \int A\left(x,y\right)v\left(x\right)dx$$

That integral transforms utilizing a kernel are analagous to matrix-vector multiplication is quite enlightening; both representing linear transformations into a more helpful basis. Even the idea of transforming into and out of problem spaces (such as Fourier above) is captured within the idea of matrix similarity;

$$\left\{\text{Transform}\right\}\left\{\text{Simpler Problem}\right\}\left\{\text{Inverse Transform}\right\} \to LDL^{-1}$$

Guaranteeing a possibly diagonal, easily solvable, or at the very least, a triangular, Schur, form.

4 Iterated Cesaro Sums

The act of taking a Cesaro average of the first n elements of a sequence can be expressed neatly as a function of its generation function;

$$f\left(x\right) = \sum_{n=0}^{\infty} x^n\cdot A\left(n\right) \to \int_0^x \frac{f\left(t\right)}{1-t}dt = \sum_{n=1}^{\infty} x^n\cdot\frac{1}{n}\sum_{k=0}^{n-1} A\left(k\right)$$

It seems plausible but likely overly cumbersome that such a transformation provides regularisation of a sum in the typical sense of Abel or Cesaro. One amusing application lies in the Dirichlet & Fejer Kernels for which we now have;

$$\sum_{n=1}^{\infty} x^n\cdot F\left(n,\theta\right) = \int_0^x \frac{\sum_{n=0}^{\infty} t^n\cdot D\left(n,\theta\right)}{1-t}dt = \int_0^x \frac{1}{1-t}\cdot\frac{t+1}{1+t^2-2t\cos\left(\theta\right)}dt = \frac{1}{2}\frac{1}{1-\cos\left(\theta\right)}\ln\left(\left|\frac{x^2-2x\cos\left(\theta\right)+1}{\left(x-1\right)^2}\right|\right)$$

Taking one more average of the Fejer Kernel eliminates almost all the oscillatory behavior,

$$\frac{1}{N}\sum_{n=1}^{N-1}F\left(n,\theta\right)=\frac{1}{1-\cos\left(\theta\right)}\frac{1}{2N}\int_{0}^{\theta}\frac{\cos\left(\frac{t}{2}\right)-\cos\left(Nt-\frac{t}{2}\right)}{\sin\left(\frac{t}{2}\right)}dt$$

$$\sum_{n=1}^{\infty}x^{n}\cdot\frac{1}{n}\sum_{k=1}^{n-1}F\left(k,\theta\right)=\frac{1}{2}\frac{1}{1-\cos\left(\theta\right)}\left[\int_{0}^{x}\frac{1}{1-t}\ln\left(\left|t^{2}-2t\cos\left(\theta\right)+1\right|\right)dt+\ln\left(\left|x-1\right|\right)^{2}\right]$$

though the resulting integral is now not so elementary. Now, returning to the original integral operation, it is quite delightful that we can use the following lemma

$$\int_{0}^{x}\int_{0}^{t_{n}}\int_{0}^{t_{n-1}}...\int_{0}^{t_{2}}\frac{1}{1-t_{n}}\frac{1}{1-t_{n-1}}...\frac{f_{1}\left(t_{1}\right)}{1-t_{1}}dt_{1}dt_{2}...dt_{n-1}dt_{n}=\int_{0}^{x}\frac{f_{1}\left(t_{1}\right)}{1-t_{1}}\frac{1}{(n-1)!}\ln\left(\frac{1-t_{1}}{1-x}\right)^{n-1}dt_{1}$$

to give a compact and elementary solution to the problem of recursively partial-averaging a given sequence–passing each generating function for the averaging into the input of the previous.

For the interested reader, it is not hard to prove the following generalization of Cauchy's iterated integral formula;

$$f_{0}\left(x\right)=f\left(x\right)\ ;\ f_{n}\left(x\right)=\int_{a}^{x}f_{n-1}\left(t\right)K\left(t\right)dt=\frac{1}{(n-1)!}\int_{a}^{x}f\left(t\right)\left(\int_{t}^{x}K\left(s\right)ds\right)^{n-1}dt$$

As produced by Leibniz integral rule;

$$f_{n}'=\frac{d}{dx}\left[\frac{1}{(n-1)!}\int_{a}^{x}f\left(t\right)\left(\int_{t}^{x}K\left(s\right)ds\right)^{n-1}dt\right]=$$

$$\frac{1}{(n-1)!}f\left(x\right)\left(\int_{x}^{x}K\left(s\right)ds\right)^{n-1}+\frac{1}{(n-2)!}\int_{a}^{x}f\left(t\right)\left(\int_{t}^{x}K\left(s\right)ds\right)^{n-2}\left[\frac{d}{dx}\int_{t}^{x}K\left(s\right)ds\right]dt=K\left(x\right)f_{n-1}$$

Plugging in $K(t)=1$ gives the classic iterated anti-derivative formula and $K(t)=\frac{1}{1-t}$ gives the desired result above.

5 Mellin-Laplace Transform Relationship

The following identity seems a bit too gorgeous to be in any way useful, but nonetheless the Reciprocal Gamma and Airy functions have a connection (as opposite sides to an inverse integral transform coin);

$$\mathcal{L}^{-1}\left\{Ai\left(t\right)\right\}=3\mathcal{M}^{-1}\left\{\frac{1}{\Gamma(s)}\right\}\left(\frac{3}{t^{3}}\right)t^{-3}$$

The key observation is to assign the Laplace transform and the Mellin Convolution integrals to be one and the same:

$$Ai\left(x\right)=\int_{0}^{\infty}e^{-xt}3\mathcal{M}^{-1}\left\{\frac{1}{\Gamma(s)}\right\}\left(\frac{3}{t^{3}}\right)\frac{1}{t^{2}}\cdot t^{-1}dt$$

$$\mathcal{M}\left\{Ai\left(x\right)\right\}=\mathcal{M}\left\{\int_{0}^{\infty}e^{-\frac{x}{y}}3\mathcal{M}^{-1}\left\{\frac{1}{\Gamma(s)}\right\}\left(3y^{3}\right)y^{2}\frac{dy}{y}\right\}$$

$$\mathcal{M}\left\{e^{-x}\right\}\cdot\mathcal{M}\left\{3\mathcal{M}^{-1}\left\{\frac{1}{\Gamma(s)}\right\}\left(3x^{3}\right)x^{2}\right\}$$

$$\Gamma(s)\cdot\mathcal{M}\left\{\mathcal{M}^{-1}\left\{\frac{1}{\Gamma(s)}\right\}\right\}\left(\frac{s+2}{3}\right)\cdot3^{-\left(\frac{s+2}{3}\right)}$$

$$\frac{\Gamma(s)}{\Gamma\left(\frac{s+2}{3}\right)}3^{-\left(\frac{s+2}{3}\right)}$$

A known result. Alternatively, one can view the convolution-Laplace relationship as the effect of multiplying by $\Gamma(s)$ in Mellin-space, for which we receive a nearly equivalent form;

$$\mathcal{L}\cdot\mathcal{M}^{-1}\cdot F=\int_{c_0-i\infty}^{c_0+i\infty}\left(\int_0^\infty e^{-tx}x^{-s}dx\right)F\left(s\right)ds=\int_{c_0-i\infty}^{c_0+i\infty}t^{(s-1)}\Gamma\left(1-s\right)F\left(s\right)ds$$

$$\rightarrow\{1-s=z\}\rightarrow\int_{c_1-i\infty}^{c_1+i\infty}t^{-z}\Gamma\left(z\right)F\left(1-z\right)dz$$

A small discussion of this method is found at MathSE No.2501698. One fun application of this relationship is;

$$\mathcal{L}\cdot\mathcal{M}^{-1}\left\{\sin\left(\frac{\pi s}{2}\right)\Gamma\left(s\right)\right\}=\int_{c_1-i\infty}^{c_1+i\infty}t^{-z}\frac{\pi}{\sin\left(\pi z\right)}\sin\left(\frac{\pi\left(1-z\right)}{2}\right)dz=$$

$$\int_{c_1-i\infty}^{c_1+i\infty}t^{-z}\frac{\pi}{2}\csc\left(\frac{\pi z}{2}\right)dz=\sum_{Res\{z=-2n\}}^{\infty}\left(-1\right)^n t^{-(-2n)}=\frac{1}{1+t^2}$$

This is perhaps the most complex way to obtain the Laplace transform of $\sin(x)$.

6 Transforming the Fransén–Robinson Constant into Series Form

The Fransén–Robinson Constant, defined from the integral of the reciprocal gamma function, can also be expressed via the well-known integral below from Ramanujan.

$$F=\int_0^\infty\frac{1}{\Gamma\left(x\right)}dx=e+\int_0^\infty\frac{e^{-x}}{\ln\left(x\right)^2+\pi^2}dx$$

$$=e+2\int_0^\infty\frac{e^{-e^{-x}}}{\left(x^2+\pi^2\right)^2}xdx-2\int_0^\infty\frac{e^{-e^{x}}}{\left(x^2+\pi^2\right)^2}xdx$$

For the first integral, a series expansion and a Laplace transform are employed;

$$\int_0^\infty\frac{e^{-e^{-x}}}{\left(x^2+\pi^2\right)^2}xdx=\frac{1}{2}\sum_{n=0}^{\infty}\frac{\left(-1\right)^n}{n!}\left(\frac{1}{\pi^2}-n\int_0^\infty\frac{e^{-nx}}{x^2+\pi^2}dx\right)$$

$$\int_0^\infty\frac{e^{-nx}}{x^2+\pi^2}dx=\int_0^\infty L^{-1}\left\{\frac{1}{t+n}\right\}\cdot L\left\{\frac{\sin\left(\pi t\right)}{\pi}\right\}dx=\frac{1}{\pi}\int_0^\infty\frac{\sin\left(\pi t\right)}{t+n}dt$$

This then becomes the Sine Integral S(x);

$$\frac{1}{2}\sum_{n=0}^{\infty}\frac{\left(-1\right)^n}{n!}\left(\frac{1}{\pi^2}-n\frac{\left(\pi-2S\left(\pi n\right)\right)\left(-1\right)^n}{2\pi}\right)$$

$$\frac{1}{2\pi^2 e}-\frac{e}{4}+\frac{1}{2\pi}\sum_{n=0}^{\infty}\frac{S\left(\pi n\right)}{\left(n-1\right)!}$$

For the second integral further Taylor expansions are used;

$$\int_0^1 e^{-e^x}e^x\frac{1}{x^2+\pi^2}dx+\int_1^\infty e^{-e^x}e^x\frac{1}{x^2+\pi^2}dx$$

Taking the first integral of the above piece;

$$\int_0^1 e^{-e^x}e^x\frac{1}{x^2+\pi^2}dx=\sum_{n=0}^{\infty}\frac{\left(-1\right)^n}{n!}\sum_{m=0}^{\infty}\frac{\left(n+1\right)^m}{m!}\int_0^1\frac{x^m}{x^2+\pi^2}dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \sum_{m=0}^{\infty} \frac{(n+1)^m}{m!} \cdot \frac{\pi^m}{\pi} \frac{1}{2} B_0\left(\frac{1}{1+\pi^2}, \left(\frac{m}{2}+\frac{1}{2}\right), \left(-\frac{m}{2}+\frac{1}{2}\right)\right)$$

Where $B_0(x,a,b)$ is the unregularized incomplete Beta function, also expressible as a power series. Alternatively, a variation of the Bell Numbers can be substituted for the outermost sum following OEIS sequence A000587;

$$\frac{-1}{2e\pi} \sum_{m=0}^{\infty} \frac{A000587_m}{m!} \pi^m B_0\left(\frac{1}{1+\pi^2}, \left(\frac{m}{2}+\frac{1}{2}\right), \left(-\frac{m}{2}+\frac{1}{2}\right)\right)$$

Now for the final integral.This sequence of splitting integrals ultimately results in a region of integration outside the origin, allowing for yet more expansions of the exponential function;

$$\int_0^1 \frac{e^{-e^{\frac{1}{x}}}}{(1+x^2\pi^2)^2} x dx = \int_e^{\infty} \frac{e^{-x}}{\left(\ln(x)^2+\pi^2\right)^2} \ln(x) \frac{1}{x} dx = \frac{-1}{2}\left(-e^{-e}\left(\frac{1}{1+\pi^2}\right) + \int_e^{\infty} e^{-x} \frac{1}{\ln(x)^2+\pi^2} dx\right)$$

$$\frac{-1}{2}\left(-e^{-e}\left(\frac{1}{1+\pi^2}\right) + \frac{1}{\pi^2}\sum_{n=0}^{\infty}\left(-\frac{1}{\pi^2}\right)^n \left(\int_e^{\infty} e^{-x}\ln(x)^{2n} dx\right)\right)$$

And finally, the complete series:

$$F = \frac{e}{2} + \frac{1}{\pi}\sum_{n=0}^{\infty} \frac{S(\pi n)}{(n-1)!} + \frac{1}{2\pi}\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\sum_{m=0}^{\infty}\frac{(n+1)^m}{m!} \cdot \pi^m \cdot B_0\left(\frac{1}{1+\pi^2}, \left(\frac{m}{2}+\frac{1}{2}\right), \left(-\frac{m}{2}+\frac{1}{2}\right)\right)$$

$$+\frac{1}{\pi^2}\sum_{n=0}^{\infty}\left(-\frac{1}{\pi^2}\right)^n \left(\int_e^{\infty} e^{-x}\ln(x)^{2n} dx\right)$$

Alternatively, in place of the Beta function, we can substitute the result of the $\int_0^1 \frac{x^m}{x^2+1} dx$ integral for an expansion based off of the reduction formula for the corresponding integral of $\int \tan(x)^m dx$;

$$F = \frac{e}{2} + \left(\frac{1}{\pi}\sum_{n=0}^{\infty}\frac{S(\pi n)}{(n-1)!}\right) + \frac{-e \operatorname{arccot}(\pi)}{\pi} + \frac{e - \left(\frac{e^{(-e)}}{2} - \frac{e^{\left(-\frac{1}{e}\right)}}{2}\right)}{\pi^2} +$$

$$\frac{1}{\pi}\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\sum_{k=1}^{\infty}\frac{(-1)^k}{\pi^{2k}}\sum_{m=k}^{\infty}\pi^{2m}(-1)^m\left(\frac{\pi}{2k}\frac{(n+1)^{(2m+1)}}{(2m+1)!} - \frac{(n+1)^{2m}}{(2m)!}\frac{1}{(2k+1)\pi}\right)$$

$$+\frac{1}{\pi^2}\sum_{n=0}^{\infty}\left(-\frac{1}{\pi^2}\right)^n \left(\int_e^{\infty} e^{-x}\ln(x)^{2n} dx\right)$$

This final integral can, following the form of the rest of the integrals, be expressed in terms of special functions,

$$\int_e^{\infty} e^{-x}\ln(x)^{2n} dx = \frac{d^{2n}}{dx^{2n}}\Gamma(x,e)\,|_{x=1}$$

for the incomplete Gamma function. Or,

$$\int_0^{\infty} e^{-x}\ln(x)^{2n} dx - \sum_{k=0}^{\infty}\frac{(-1)^k}{k!}\int_0^e x^k \ln(x)^{2n} dx$$

$$= \Gamma^{(2n)}(1) - \sum_{k=0}^{\infty}\frac{(-1)^k}{k!}\left(\frac{(2n)!}{(k+1)^{(2n+1)}} - \sum_{m=0}^{\infty}\frac{(-1)^m (k+1)^m}{m!}\frac{(-1)^{(m+1)}}{m+2n+1}\right)$$

for the ordinary gamma function.